Computer Science practice paper 2: answers and marks

How to mark your paper. Go through one question at a time. Read the answer, then check your own answer against "How the marks are given". Give yourself the mark if your answer means the same thing, even if you used different words.

Write your mark next to each question, then add them up. Read the "Watch out" boxes even when you got the marks: they're the mistakes that cost people marks in the real exam.

1(a)

1 mark

State what is meant by abstraction.

The answer

Abstraction means removing the details that aren't needed, so you can focus on the important parts of the problem.

How the marks are given

- 1 mark: removing or ignoring unnecessary detail (or focusing on what's important).

Watch out

- Don't mix it up with decomposition, which is breaking a problem down.

1(b)

1 mark

State what is meant by decomposition.

The answer

Decomposition means breaking a big problem down into smaller problems that are easier to solve.

How the marks are given

- 1 mark: breaking a problem into smaller parts (sub-problems).

1(c)

1 mark

Here is a list of numbers: 3, 8, 12, 15, 21, 27, 30. Explain why a binary search can be used on this list.

The answer

Because the list is in order (sorted).

How the marks are given

- 1 mark: the list is sorted, or in order.

Watch out

- A binary search only works on a sorted list. That's the key point.

1(d)

3 marks

Show the steps a binary search would take to find the number 21 in the list 3, 8, 12, 15, 21, 27, 30.

The answer

Look at the middle item, 15. 21 is bigger than 15, so throw away 15 and everything before it, leaving 21, 27, 30. The middle of those is 27. 21 is smaller than 27, so throw away 27 and 30, leaving 21. The middle item is 21, so it's found.

How the marks are given

- 1 mark: you compare 21 with the middle item, 15.
- 1 mark: you get rid of the half that can't have 21 in it (15 and everything before it).
- 1 mark: you compare with 27, then find 21.

Watch out

- Always start at the middle item, not the first one. Starting at the beginning and checking each item is a linear search.

1(e)

2 marks

A bubble sort is used to put the list 5, 2, 9, 1 into ascending order. Write the list as it is after the first pass.

The answer

2, 5, 1, 9. Swap 5 and 2 (2, 5, 9, 1). Leave 5 and 9. Swap 9 and 1 (2, 5, 1, 9).

How the marks are given

- 2 marks: 2, 5, 1, 9.
- 1 mark if one swap is right but the final list is wrong.

Watch out

- One pass means going through the list once, comparing each pair. It isn't fully sorted yet after one pass: 1 is still in the wrong place.

2(a)

2 marks

```
total = 0
for i = 1 to 4
    total = total + i * 2
next i
print(total)
```

Give the value of total after each time round the loop, and the value that is printed.

The answer

total is 2, then 6, then 12, then 20. It prints 20.

How the marks are given

- 1 mark: the values 2, 6 and 12.
- 1 mark: it prints 20.

Watch out

- Work out i * 2 first, then add it on. Each time round, i goes up by 1: 1, 2, 3, 4.

2(b)

1 mark

State the type of iteration used in this code.

The answer

Count-controlled iteration (a for loop).

How the marks are given

- 1 mark: count-controlled (or definite).

Watch out

- A while loop is condition-controlled. This one is a for loop, so it runs a set number of times.

2(c)

2 marks

State the most suitable data type for each of these values: 15, 3.5, "Yes", True.

The answer

15 is an integer, 3.5 is a real (also called a float), "Yes" is a string, and True is a Boolean.

How the marks are given

- 2 marks: all four right.
- 1 mark: two or three right.

Watch out

- "Yes" has speech marks, so it's a string, not a Boolean.

2(d)

1 mark

What is the result of 17 MOD 5?

The answer

2. 17 divided by 5 is 3 with 2 left over, and MOD gives the remainder.

How the marks are given

- 1 mark: 2.

Watch out

- MOD gives the remainder. DIV gives the whole-number part (17 DIV 5 is 3).

2(e)

3 marks

Write an algorithm that asks the user to enter a password. It should keep asking until they enter letmein, then print Access granted. You can use OCR Exam Reference Language or Python.

The answer

```
password = input("Enter the password")
while password != "letmein"
   password = input("Enter the password")
endwhile
print("Access granted")
```

How the marks are given

- 1 mark: asks the user to input the password.
- 1 mark: a loop that keeps going until the password is letmein (for example, while password != "letmein").
- 1 mark: prints Access granted after the loop, once it's right.

Watch out

- An if statement on its own only checks once. The question says keep asking, so you need a loop.
- Python is fine: while password != "letmein": then the next line indented.

3(a)

2 marks

A program asks for a student's age. The age must be from 11 to 18. Name a validation check that could be used, and describe what it would check.

The answer

A range check. It checks the age is 11 or more and 18 or less.

How the marks are given

- 1 mark: range check.
- 1 mark: it checks the age is between 11 and 18 (11 or more and 18 or less).

Watch out

- A presence check or a type check would only get the first mark if you described what it checks.

3(b)

3 marks

Give one example of each type of test data for the age: normal, boundary and erroneous.

The answer

Normal: 14. Boundary: 11 (or 18). Erroneous: "fifteen" (the wrong type of data).

How the marks are given

- 1 mark: a normal value, such as 12 to 17.
- 1 mark: a boundary value: 11 or 18.
- 1 mark: an erroneous value: the wrong type of data, such as letters.

Watch out

- Erroneous means the wrong type of data, like text when a number is wanted. A number like 25 is invalid, not erroneous.

3(c)

1 mark

Give one way a programmer can make their code easier to maintain.

The answer

Add comments to explain what the code does.

How the marks are given

- 1 mark: any one: comments, sensible variable names, indentation, or splitting the code into subprograms (functions or procedures).

4(a)

2 marks

A | B | P: 0 0; 0 1; 1 0; 1 1

P = A AND NOT B. Write the value of P for each row of the table, from top to bottom.

The answer

0, 0, 1, 0.

How the marks are given

- 2 marks: all four right.
- 1 mark: three right.

Watch out

- Do NOT B first (it flips B), then AND it with A. P is only 1 when A is 1 and B is 0.

4(b)

1 mark

Name the logic gate that outputs 1 only when both of its inputs are 1.

The answer

AND.

How the marks are given

- 1 mark: AND.

Watch out

- OR outputs 1 if either input is 1. NOT has only one input.

5(a)

2 marks

Give two differences between a compiler and an interpreter.

The answer

A compiler translates the whole program in one go, but an interpreter translates and runs it one line at a time. A compiler makes an executable file you can run without the compiler, but an interpreter has to be there every time the program runs.

How the marks are given

- 1 mark for each difference, up to 2 marks. Each one must say what both do.
- Differences that count: all at once or line by line; makes an executable file or doesn't; reports all the errors at the end or stops at the first error; compiled programs run faster.

Watch out

- Saying only what a compiler does isn't a difference. Compare both.

5(b)

2 marks

Name one feature of an IDE (integrated development environment) and say how it helps a programmer.

The answer

Syntax highlighting: it colours the code, so mistakes like a missing bracket are easier to spot.

How the marks are given

- 1 mark: name a feature (editor, syntax highlighting, error diagnostics, debugger or breakpoints, run-time environment, auto-complete, translator).
- 1 mark: say how that feature helps.

Watch out

- The way it helps must match the feature you named.

Total: 30 marks. A rough guide for this practice paper: 22 or more is a strong result. 15 to 21 is solid, with some gaps to fix. Under 15 means going back over these topics in the notebook first. Whatever you got, write down the questions where you lost marks: those are what to revise next.